



Problems (Integration)

1. Evaluate the following

- a. $\int 7^{2x+3} dx$
- b. $\int \sqrt{7x+9} dx$
- c. $\int (2x+5)(x^2+5x)^8 dx$
- d. $\int \frac{\sin(\ln x)}{x} dx$
- e. $\int x\sqrt{4-x} dx$
- f. $\int (\ln x)^3 dx$
- g. $\int \left(\frac{\ln x}{x}\right)^2 dx$

- 2. Find the area under the curve $y = \sqrt{x}$ between the origin and (4,0)
- 3. Find the area of the region enclosed by $y = x$, $y = 3x$ and $x = 2$